

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

 show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 129

Does R2 show up as an OSPF neighbor on R1? Yes

Does R2 show up as an OSPF neighbor on R3? No

What does this information tell you?

 Only interfaces configured as non-passive (no passive-interface) can send Hello packets and establish neighbor relationships. R2 has re-established the neighbor relationship with R1, indicating that the interface S0/0/0 of R2 has returned to normal operation for sending and receiving data. While there is no neighbor relationship established with R3, indicating that the S0/0/1 interface connected to R3 is still in a passive shielding state, preventing R2 from directly notifying its routing information to R3.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

 R2(config)# router ospf 1

 R2(config-router)# no passive-interface s0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

 65 (serial link cost 64 + Ethernet cost 1)

Is R2 listed as an OSPF neighbor to R3? Yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

The default reference bandwidth for OSPF is 100 Mbps, which results in all 100 Mbps and above high-speed links (such as gigabit and 10-gigabit interfaces) having an equal cost of 1. Modifying this value enables OSPF to accurately distinguish and calculate the actual costs of different high-speed links, thereby making the optimal routing decisions.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

At this point, the bandwidth of both S0/0/0 and S0/0/1 interfaces on R1 has been changed to 128 kbps, and the interface overhead has become 781.

The cost of reaching 192.168.3.0/24 is 782. The calculation method is: the link cost of the serial link S0/0/1 on R1 (781) + the cost of the gigabit local area network interface on R3 (1).

The cost of reaching 192.168.23.0/30 is 845. The calculation method is: the cost of S0/0/1 interface on R1 (781) + the default cost of the target network segment at the opposite end interface (64).

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

1562. Because at this point, the overhead for all the serial interfaces in the topology has become 781. Whether reaching this network via R1 through R2, or via R3, the total overhead is the sum of the overheads of the two serial interfaces of the route (781 + 781 = 1562).

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

___ At this time, the total cost via R2 as an alternative route was 1563 (the cost from R1 to R2 was 781 + the cost from R2 to R3 was 781 + the local area network cost of R3 was 1). Because after manually configuring ip ospf cost 1565 on the S0/0/1 interface of R1, the cost for directly reaching this local area network through R3 became 1566 (1565 + 1). Since OSPF always selects the path with the lowest cost, the routing strategy will switch to using R2 as the alternative route. _____

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

___ The Router ID is a unique identifier for each router in an OSPF domain, used to maintain adjacency relationships. Assigning Router IDs manually and in a planned manner helps prevent routing confusion caused by ID conflicts and enables administrators to identify topology nodes more quickly during troubleshooting. _____

2. Why is the DR/BDR election process not a concern in this lab?

___ In the experiment, all backbone connections between routers used serial point-to-point links, while DR/BDR elections occurred only on the multi-access network. Although the PCs were connected via Gigabit Ethernet, there was no election competition because there was only one router in each LAN. _____

3. Why would you want to set an OSPF interface to passive?

___ Setting it in passive mode can prevent the interface from sending OSPF Hello packets and stop the establishment of neighbor relationships. This not only eliminates unnecessary multicast traffic on the user-end LAN, but also prevents malicious devices from accessing the interface to spy on or disrupt the internal routing information. _____

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

___ OSPF introduced the concept of "Area" (for example, Area 0 is the backbone area). This hierarchical area design can limit the scope of routing updates and SPF calculations to a smaller range, effectively controlling the size of the routing table, thereby enabling the construction of extremely large and highly scalable networks. _____

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

___ router ospf 1 network 10.0.0.0 0.255.255.255 area 0 _____

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

___ The router will select the routes provided by EIGRP. This is because when making decisions between different protocols, the router only considers the administrative distance (AD). The internal AD value of EIGRP is 90, that of OSPF is 110, and that of RIP is 120. EIGRP has the lowest AD value, so it has the highest priority. At this time, the sizes of the Metrics of each protocol itself are ignored. _____